

Qwen3.5 Usage Guide

Qwen3.5 is a [multimodal](#) mixture-of-experts model featuring a gated delta networks architecture with 397B total parameters and 17B active parameters.

This guide covers how to efficiently deploy and serve the model across different hardware configurations and workload profiles using vLLM.

Installing vLLM

You can either install vLLM from pip or use the pre-built Docker image. 

Pip Install

NVIDIA

```
uv venv
source .venv/bin/activate
uv pip install -U vllm --torch-backend=auto
```



AMD

Note: The vLLM wheel for ROCm requires Python 3.12, ROCm 7.0, and glibc >= 2.35.

If your environment does not meet these requirements, please use the Docker-based setup as described below.

Supported GPUs: MI300X, MI325X, MI355X.

```
uv venv --python 3.12
source .venv/bin/activate
uv pip install vllm --extra-index-url https://wheels.vllm.ai/rocm
```



Docker

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NVIDIA

```
docker run --gpus all \
  -p 8000:8000 \
  --ipc=host \
  -v ~/.cache/huggingface:/root/.cache/huggingface \
  vllm/vllm-openai Qwen/Qwen3.5-397B-A17B \
  --tensor-parallel-size 8 \
  --reasoning-parser qwen3 \
  --enable-prefix-caching
```

(See detailed deployment configurations below)

For Blackwell GPUs, use `vllm/vllm-openai:cu130-nightly`

AMD

```
docker run --device=/dev/kfd --device=/dev/dri \
  --security-opt seccomp=unconfined \
  --group-add video \
  --ipc=host \
  -p 8000:8000 \
  -v ~/.cache/huggingface:/root/.cache/huggingface \
  vllm/vllm-openai-rocm:latest \
  Qwen/Qwen3.5-397B-A17B-FP8 \
  --tensor-parallel-size 8 \
  --reasoning-parser qwen3 \
  --enable-prefix-caching
```

Running Qwen3.5

The configurations below have been verified on 8x H200 GPUs and 8x MI300X/MI355X GPUs.



Tip

We recommend using the official FP8 checkpoint [Qwen/Qwen3.5-397B-A17B-FP8](#) for optimal serving efficiency.

Throughput-Focused Serving

Text-Only

For maximum text throughput under high concurrency, use `--language-model-only` to skip loading the vision encoder and free up memory for KV cache as well as enabling Expert Parallelism.

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```
vllm serve Qwen/Qwen3.5-397B-A17B-FP8 \
  -dp 8 \
  --enable-expert-parallel \
  --language-model-only \
  --reasoning-parser qwen3 \
  --enable-prefix-caching
```

Multimodal

For [multimodal workloads](#), use `--mm-encoder-tp-mode data` for data-parallel vision encoding and `--mm-processor-cache-type shm` to efficiently cache and transfer preprocessed [multimodal inputs](#) in shared memory.

```
vllm serve Qwen/Qwen3.5-397B-A17B-FP8 \
  -dp 8 \
  --enable-expert-parallel \
  --mm-encoder-tp-mode data \
  --mm-processor-cache-type shm \
  --reasoning-parser qwen3 \
  --enable-prefix-caching
```



Tip

To enable tool calling, add `--enable-auto-tool-choice --tool-call-parser qwen3_coder` to the serve command.

Latency-Focused Serving

For latency-sensitive workloads at low concurrency, enable MTP-1 speculative decoding and disable prefix caching. MTP-1 reduces time-per-output-token (TPOT) with a high acceptance rate, at the cost of lower [throughput](#) under load.



Note

MTP-1 speculative decoding for AMD GPUs is under development. 

```
vllm serve Qwen/Qwen3.5-397B-A17B-FP8 \
  --tensor-parallel-size 8 \
  --speculative-config '{"method": "mtp", "num_speculative_tokens": 1, "num_speculative_steps": 1}' \
  --reasoning-parser qwen3
```

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GB200 Deployment



Tip

We recommend using the NVFP4 checkpoint [nvidia/Qwen3.5-397B-A17B-NVFP4](#) for optimal serving efficiency.

You can also deploy the model across 4 GPUs on a GB200 node, using the similar base configuration as H200.

```
vllm serve nvidia/Qwen3.5-397B-A17B-NVFP4 \  
-dp 4 \  
--enable-expert-parallel \  
--language-model-only \  
--reasoning-parser qwen3 \  
--enable-prefix-caching
```



MI355X Deployment

You can also deploy the model across 2 GPUs on a MI355X node, using the similar base configuration as above.

```
vllm serve Qwen/Qwen3.5-397B-A17B-FP8 \  
-tp 2 \  
--enable-expert-parallel \  
--language-model-only \  
--reasoning-parser qwen3 \  
--enable-prefix-caching
```



Configuration Tips

Prefix Caching: Prefix caching for Mamba cache "align" mode is currently experimental.

Please report any issues you may observe.

Multi-token Prediction: MTP-1 reduces per-token latency but degrades text throughput under high concurrency

because speculative tokens consume KV cache capacity, reducing effective batch size.

Depending on your use case,

you may adjust `num_speculative_tokens`:




```
--model Qwen/Qwen3.5-397B-A17B \
--dataset-name random \
--random-input-len 2048 \
--random-output-len 512 \
--num-prompts 1000 \
--request-rate 20
```

Consume the OpenAI API Compatible Server

```
import time
from openai import OpenAI

client = OpenAI(
    api_key="EMPTY",
    base_url="http://localhost:8000/v1",
    timeout=3600
)

messages = [
    {
        "role": "user",
        "content": [
            {
                "type": "image_url",
                "image_url": {
                    "url": "https://ofasys-multimodal-wlcb-3-toshanghai.oss-
accelerate.aliyuncs.com/wpf272043/keepme/image/receipt.png"
                }
            },
            {
                "type": "text",
                "text": "Read all the text in the image."
            }
        ]
    }
]

start = time.time()
response = client.chat.completions.create(
    model="Qwen/Qwen3.5-397B-A17B",
    messages=messages,
    max_tokens=2048
)
print(f"Response costs: {time.time() - start:.2f}s")
print(f"Generated text: {response.choices[0].message.content}")
```

Processing Ultra-Long Texts

Qwen3.5 natively supports context lengths of up to 262,144 tokens

For long-horizon tasks where the total length
(including both input and output)

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exceeds this limit,
we recommend using RoPE scaling techniques
to handle long texts effectively,
e.g., YaRN. you can override the `rope_parameters` in your running script.
Refer to [Qwen3.5-397B-A17B](#) for more details..

```
export VLLM_ALLOW_LONG_MAX_MODEL_LEN=1
VLLM_ALLOW_LONG_MAX_MODEL_LEN=1 vllm serve ... --hf-overrides '{"text_config":
{"rope_parameters": {"mrope_interleaved": true, "mrope_section": [11, 11, 10],
"rope_type": "yarn", "rope_theta": 10000000, "partial_rotary_factor": 0.25,
"factor": 4.0, "original_max_position_embeddings": 262144}}}' --max-model-len
1010000
```

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